

Growth and Stem-Cell Data: Evidence Before Claims

LAUNCH • Science • 40 minutes

I can interpret growth and stem-cell data, calculate changes or proportions, and evaluate application claims using evidence and biological limitations.

Prepare before pupils arrive

Setup: Prepare only fictional/anonymised datasets with legible axes, sample sizes and control information. Check calculations and separate marker evidence, function, safety, efficacy and ethical acceptability.

Safety: Keep examples non-diagnostic and emotionally safe. Do not solicit personal medical histories, promise outcomes or ask pupils to defend a disclosed condition; signpost approved support where needed.

Equipment: Fictional/anonymised growth chart, fictional stem-cell results table, calculator, ruler, command-word guide and claim-strength cards.

Safety: Use no personal growth or medical data. This is not a diagnostic or treatment activity; use supplied fictional cases, avoid personal disclosure and route health concerns through approved school support.

Fallback: Use enlarged pre-plotted graphs, table overlays, calculator prompts, oral choice cards and exact pupil-directed scribing. Retain the calculation and evidence-limit requirements.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation • Paper 1 • Topic 2 • specification points 2.5–2.9, growth data, differentiation and stem-cell applications.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 3 min	State one limitation of interpreting a single growth measurement. Write the equation for percentage change.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	percentile — position in a reference distribution efficacy — how well an intervention works inference — reasoned interpretation For a fictional growth chart, read age, measurement and unit, then describe direction and change using values rather than labels such as normal or abnormal. Calculate a change or percentage change and check the denominator.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	For a fictional stem-cell dataset, identify groups, sample sizes, measured outcome and time point before calculating a proportion or difference. Use the result to make a bounded claim: the intervention increased or decreased the measured outcome under the study conditions.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each statement to directly supported, reasonable inference or unsupported overclaim before composing the evaluation. Read the dataset and one candidate statement.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Complete a GCSE application pack containing a growth-chart interpretation, two shown calculations and an evidence-led evaluation of a fictional stem-cell claim. Record: A value-and-unit growth description, shown calculation, stem-cell data comparison, biological explanation, limitation and supported judgement ready to feed W11 ethics work.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.

Stage / total time	Teaching move	Adult support / response
Exit · 4 min	Use one value to support a stem-cell application claim and one limitation to bound it. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: The measurements show a broadly consistent relative growth pattern across these time points.

Correct. The chart supports a pattern description, not a diagnosis or direct claim about individual cell processes.

Misconception repair

Repaired. The judgement uses effect size, design, uncertainty and biological plausibility without promising a universal outcome.

Guided checkpoint

Correct. The chart supports a pattern description, not a diagnosis or direct claim about individual cell processes. Choose the answer restricted to measured position across the supplied time points.

Measured increase

DIRECTLY SUPPORTED. Directly supported: both values, unit and interval come from the dataset. Look for a statement that simply reports measured values.

Marker percentage

DIRECTLY SUPPORTED. Directly supported: the percentage describes this measured culture outcome only. Do not add a treatment result that the culture did not measure.

Cell-process link

REASONABLE INFERENCE. Reasonable inference: biological knowledge explains the pattern while the wording admits it was not directly observed. This connects evidence to a plausible process without saying it was measured.

Potential repair

REASONABLE INFERENCE. Reasonable inference: the benefit is conditional and does not claim proven clinical success. Look for conditional language and missing evidence stated openly.

One-point diagnosis

UNSUPPORTED OVERCLAIM. Unsupported: one chart point cannot establish a diagnosis or cause. The claim goes beyond a monitoring graph's evidence.

Guaranteed cure

UNSUPPORTED OVERCLAIM. Unsupported: culture markers do not establish patient safety, efficacy or universal outcome. Ask whether patients, harms and lasting functional outcomes were measured.

Single-cause claim

UNSUPPORTED OVERCLAIM. Unsupported without controlled evidence: association alone does not exclude other explanations. Check the control group and confounding variables before claiming causation.

Model limitation

Simplified datasets omit individual variation, full methods, longer-term outcomes and many confounders. Percentile charts and cell markers are useful evidence tools but cannot diagnose or establish a treatment's complete safety and effectiveness alone.

Independent evidence

A value-and-unit growth description, shown calculation, stem-cell data comparison, biological explanation, limitation and supported judgement ready to feed W11 ethics work.

Supported: Read two graph values, calculate one change and one worked percentage, then select and justify a bounded conclusion. Use highlighted axes, calculator key sequence and stems: The data show __. This could mean __, but they do not show __.

Standard: Interpret the growth pattern, calculate change and cell percentage, then evaluate the fictional application using one benefit and two evidence limitations. Work in DESCRIBE → CALCULATE → EXPLAIN → EVALUATE order and quote at least two values.

Stretch: Compare groups using quantitative evidence, examine variation or control quality, and reach a conditional judgement that separates efficacy, safety and ethics. Identify what further data would reduce uncertainty and explain why; do not decide the W11 ethical question yet.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W10L3_Growth_Stem_Cell_Data_Application_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.